

The Rotation Orbit Tower and a Proof of $h_m \geq 0$ for $d = 2$

Seed 4, Layer 2 — Warnaar positivity project

Seed 4 (Layer 2, Round 2)

2026-07-04

Abstract

We prove two results toward the core bottleneck of the positivity conjecture for $Q_{n,c}(q)$ ($r = 3$, $d \not\equiv 0 \pmod{3}$), namely the conjecture $h_m := (q^\ell; q^\ell)_m g_m \geq 0$ with $g_m = P_m/(q^3; q^3)_m$ and $\ell = \gcd(d, 3) = 1$. First, an *orbit-tower reduction* valid for all d with $3 \nmid d$: h_m -positivity is equivalent to coefficientwise nonnegativity of an explicit tower $H_m = U(q^m)H_{m-1}$, $H_0 = \mathbf{1}$, where $U(x)$ is a fixed $N \times N$ polynomial matrix, $N = \frac{(d+1)(d+2)}{6}$, indexed by the $\mathbb{Z}/3$ -rotation orbits of profiles. The exact divisibility making U polynomial follows from a two-line congruence for the cyclic earth-mover distance. Second, for $d = 2$ we solve the tower in closed form: h_m equals a finite Rogers–Ramanujan (MacMahon–Schur) polynomial, hence is manifestly nonnegative. The proof uses only the q -Pascal and absorption identities.

1 Setup

Fix $r = 3$ and $d \geq 1$ with $3 \nmid d$. A *profile* is a composition $c = (c_0, c_1, c_2)$ of d into three nonnegative parts; there are $\binom{d+2}{2}$ of them. The cyclic (clockwise) earth-mover distance is

$$\text{EMD}(c, c') = 3 \max(0, \alpha, \beta) - \alpha - \beta, \quad \alpha = c'_1 - c_1, \quad \beta = c_0 - c'_0.$$

Define $P_m(c)$ as the generating function of weighted profile paths,

$$P_0(c) = 1, \quad P_m(c) = \sum_{c'} q^{m \text{EMD}(c, c')} P_{m-1}(c'),$$

so that $P_m(c) = \sum_{\gamma} q^{\sum_{k=1}^m k \text{EMD}(c^k, c^{k-1})}$ over paths $\gamma = (c^0, \dots, c^m = c)$. The Layer-1 results (Adjugate Monomial Theorem and its consequences, all verified) express the core quantity as

$$h_m(c) = \frac{(q; q)_m}{(q^3; q^3)_m} P_m(c) = \frac{P_m(c)}{\prod_{k=1}^m (1 + q^k + q^{2k})},$$

using $(1 - q^{3k})/(1 - q^k) = 1 + q^k + q^{2k}$. The bottleneck conjecture is that $h_m(c)$ is a polynomial with nonnegative coefficients.

Let $\rho(c) = (c_2, c_0, c_1)$ denote cyclic rotation of profiles. Directly from the formula, $\text{EMD}(\rho a, \rho b) = \text{EMD}(a, b)$; hence by induction $P_m(\rho c) = P_m(c)$ and h_m is constant on ρ -orbits. Since $3 \nmid d$, no profile is ρ -fixed, so all orbits have size 3 and there are $N = \binom{d+2}{2}/3 = \frac{(d+1)(d+2)}{6}$ of them.

2 The orbit tower

Lemma 1 (mod-3 congruence for EMD). *For all profiles a, b of d ,*

$$\text{EMD}(a, b) \equiv b_0 - a_0 + a_1 - b_1 \pmod{3}, \quad \text{EMD}(a, \rho b) - \text{EMD}(a, b) \equiv d \pmod{3}.$$

Proof. Since $\text{EMD} = 3 \max(0, \alpha, \beta) - \alpha - \beta \equiv -\alpha - \beta = (a_1 - b_1) + (b_0 - a_0) \pmod{3}$, the first congruence holds. With $\rho b = (b_2, b_0, b_1)$,

$$[(\rho b)_0 - a_0 + a_1 - (\rho b)_1] - [b_0 - a_0 + a_1 - b_1] = (b_2 - b_0) - (b_0 - b_1) = b_0 + b_1 + b_2 - 3b_0 \equiv d \pmod{3}. \quad \square$$

Theorem 2 (Orbit-tower reduction, all $3 \nmid d$). *Choose orbit representatives $a_1, \dots, a_N$, and for each pair (i, j) set $E_t(i, j) = \text{EMD}(a_i, \rho^t a_j)$, $t = 0, 1, 2$. Then:*

1. *The exponents E_0, E_1, E_2 are pairwise distinct modulo 3; hence $x^{E_0} + x^{E_1} + x^{E_2}$ is divisible by $1 + x + x^2$ in $\mathbb{Z}[x]$, and*

$$U_{ij}(x) := \frac{x^{E_0(i,j)} + x^{E_1(i,j)} + x^{E_2(i,j)}}{1 + x + x^2} \in \mathbb{Z}[x].$$

2. *The vector $H_m = (h_m(a_1), \dots, h_m(a_N))$ satisfies*

$$H_m = U(q^m) H_{m-1}, \quad H_0 = (1, \dots, 1)^\top.$$

Consequently $h_m \geq 0$ (all c , all m) if and only if the tower H_m is coefficientwise nonnegative for all m .

Proof. (1) By Lemma ??, $E_t \equiv E_0 + td \pmod{3}$; since $3 \nmid d$ these are the three residues in some order. If ω is a primitive cube root of unity, $\omega^{E_0} + \omega^{E_1} + \omega^{E_2} = 1 + \omega + \omega^2 = 0$, so both primitive cube roots of unity are roots of $x^{E_0} + x^{E_1} + x^{E_2}$, which is therefore divisible by $1 + x + x^2$.

- (2) Group the P -recursion by orbits, using ρ -invariance of P_{m-1} :

$$P_m(a_i) = \sum_{j=1}^N \left(\sum_{t=0}^2 q^{m E_t(i,j)} \right) P_{m-1}(a_j).$$

Divide both sides by $\prod_{k \leq m} (1 + q^k + q^{2k})$; the factor $1 + q^m + q^{2m}$ absorbs into the bracket by part (1) with $x = q^m$, giving the stated system for H_m . The base case is $P_0 = 1$. $\square$

Remark 3. Lemma ?? turns the equidistribution phenomenon observed computationally in earlier layers (Seed 7) into a proof. Theorem ?? replaces the infinite family of statements “ $(q; q)_m P_m / (q^3; q^3)_m \geq 0$ ” by nonnegativity of a tower generated by one *fixed, explicitly computable* polynomial matrix.

3 Solution of the tower for $d = 2$

For $d = 2$ there are $N = 2$ orbits, with representatives $a_1 = (2, 0, 0)$ and $a_2 = (1, 1, 0)$. Direct computation of the twelve values $\text{EMD}(a_i, \rho^t a_j)$ gives the exponent triples

$$\{E_t(1, 1)\} = \{0, 2, 4\}, \quad \{E_t(1, 2)\} = \{E_t(2, 1)\} = \{1, 2, 3\}, \quad \{E_t(2, 2)\} = \{0, 1, 2\},$$

so, since $(1 + x^2 + x^4) = (1 + x + x^2)(1 - x + x^2)$,

$$U(x) = \begin{pmatrix} 1 - x + x^2 & x \\ x & 1 \end{pmatrix}.$$

Recall the Gaussian binomial $\begin{bmatrix} m \\ j \end{bmatrix}_q$ and define the finite Rogers–Ramanujan (MacMahon–Schur) polynomials

$$A_m = \sum_{j \geq 0} q^{j^2+j} \begin{bmatrix} m \\ j \end{bmatrix}_q, \quad B_m = \sum_{j \geq 0} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q, \quad C_m = \sum_{j \geq 0} q^{j^2+2j} \begin{bmatrix} m \\ j \end{bmatrix}_q.$$

Theorem 4 ($d = 2$ solved). *For $d = 2$ and all $m \geq 0$,*

$$h_m(c) = A_m \text{ if } c \in \{(2, 0, 0), (0, 2, 0), (0, 0, 2)\}, \quad h_m(c) = B_m \text{ if } c \in \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}.$$

In particular $h_m(c)$ has nonnegative coefficients for all c and m .

The proof rests on three elementary identities. We use the q -Pascal rule $\begin{bmatrix} m \\ j \end{bmatrix}_q = \begin{bmatrix} m-1 \\ j \end{bmatrix}_q + q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q$ and the absorption identities $(1-q^j) \begin{bmatrix} n \\ j \end{bmatrix}_q = (1-q^n) \begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_q$ and $(1-q^{n-j}) \begin{bmatrix} n \\ j \end{bmatrix}_q = (1-q^n) \begin{bmatrix} n-1 \\ j \end{bmatrix}_q$.

Identity 5. $B_m - B_{m-1} = q^m A_{m-1}$.

Proof. By q -Pascal, $B_m - B_{m-1} = \sum_j q^{j^2} q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q$. Substituting $j = i+1$: $q^m \sum_i q^{i^2+i} \begin{bmatrix} m-1 \\ i \end{bmatrix}_q = q^m A_{m-1}$. $\square$

Identity 6. $A_m - A_{m-1} = q^{m+1} C_{m-1}$.

Proof. By q -Pascal, $A_m - A_{m-1} = \sum_j q^{j^2+j} q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q = q^m \sum_i q^{(i+1)^2} \begin{bmatrix} m-1 \\ i \end{bmatrix}_q = q^{m+1} C_{m-1}$. $\square$

Identity 7 (key lemma). $q C_n = B_n - (1 - q^{n+1}) A_n$ for all $n \geq 0$.

Proof. Compute

$$B_n - A_n + q^{n+1} A_n - q C_n = \sum_j \begin{bmatrix} n \\ j \end{bmatrix}_q q^{j^2} \left((1 - q^j) - q^{2j+1} (1 - q^{n-j}) \right).$$

Apply the first absorption identity to the term $(1 - q^j) \begin{bmatrix} n \\ j \end{bmatrix}_q$ and the second to $(1 - q^{n-j}) \begin{bmatrix} n \\ j \end{bmatrix}_q$:

$$= (1 - q^n) \left(\sum_j q^{j^2} \begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_q - \sum_j q^{j^2+2j+1} \begin{bmatrix} n-1 \\ j \end{bmatrix}_q \right) = (1 - q^n) \left(\sum_i q^{i^2+2i+1} \begin{bmatrix} n-1 \\ i \end{bmatrix}_q - \sum_j q^{j^2+2j+1} \begin{bmatrix} n-1 \\ j \end{bmatrix}_q \right) = 0,$$

where the first sum was reindexed by $j = i + 1$. (For $n = 0$ all three of A_0, B_0, C_0 equal 1 and the identity reads $q = 1 - (1 - q)$.) $\square$

Proof of Theorem ??. By Theorem ?? it suffices to show that (A_m, B_m) satisfies the $d = 2$ tower, i.e. with $x = q^m$,

$$A_m = (1 - x + x^2)A_{m-1} + xB_{m-1}, \quad B_m = xA_{m-1} + B_{m-1},$$

with base case $A_0 = B_0 = 1 = H_0$. The B -equation is Identity ??. For the A -equation,

$$(1 - q^m + q^{2m})A_{m-1} + q^m B_{m-1} = A_{m-1} + q^m (B_{m-1} - (1 - q^m)A_{m-1}) = A_{m-1} + q^m \cdot qC_{m-1} = A_m,$$

using Identity ?? with $n = m - 1$ and then Identity ??. Since the tower has a unique solution given H_0 , we conclude $H_m = (A_m, B_m)$, i.e. $h_m(a_1) = A_m$, $h_m(a_2) = B_m$, and by ρ -invariance the same holds on the full orbits. Nonnegativity is manifest from the defining sums. $\square$

Example ($m = 2$). $A_2 = 1 + q^2 + q^3 + q^6$, $B_2 = 1 + q + q^2 + q^4$. Check the tower with $x = q^2$: $B_2 = q^2 A_1 + B_1 = q^2(1 + q^2) + (1 + q) \checkmark$; $A_2 = (1 - q^2 + q^4)(1 + q^2) + q^2(1 + q) = (1 + q^6) + (q^2 + q^3) \checkmark$. Also $\prod_{k \leq 2} (1 + q^k + q^{2k}) \cdot A_2 = P_2((2, 0, 0))$, verified against the direct path enumeration.

4 Status for $d = 4$ and outlook

For $d = 4$ the tower is 5×5 . Computationally, three of the five orbit values admit bounded fermionic forms of Warnaar's Conjecture-2 shape

$$H_m = \sum_{n,j} q^{n^2 - nj + j^2 + an + bj} \begin{bmatrix} m \\ n \end{bmatrix}_q \begin{bmatrix} 2n \\ j \end{bmatrix}_q,$$

namely the orbits of $(1, 1, 2)$, $(0, 0, 4)$, and $(0, 3, 1)$ (verified for $m \leq 5$). The two remaining orbits, of $(0, 1, 3)$ and $(0, 2, 2)$, provably do not fit a single sum of this shape, and an exhaustive search over pairs of such sums also failed; these profiles lie outside the family covered by Warnaar's Conjecture 2 and finding their fermionic forms is open. The $d = 2$ proof provides the template: once closed forms are found for all five orbits, the tower identities should again reduce to q -Pascal manipulations.